

TUTORIAL 4: INTEGRATION

SECTION A

1. Evaluate $\int_2^4 \left(1 + \frac{5}{x}\right)(x-2)^2 dx$.

2. Evaluate the following:

a) $\int_{-2}^4 (t+1)(t^2+3t-5) dt$

b) $\int_0^{\frac{\pi}{2}} \sec^2 \frac{\theta}{2} d\theta$

3. Determine $\int \frac{(1-\sqrt{x})^2}{x^2} dx$.

4. Determine $\int x^{\frac{1}{4}}(1-x)^2 dx$.

5. Evaluate $\int_1^{\frac{\pi}{2}} \left(e^{3x} + \sin 3x - \frac{2}{x}\right) dx$.

6. Evaluate the following:

a) $\int_0^{\frac{9\pi}{4}} \frac{2 \tan \theta}{\sin 2\theta} d\theta$

b) $\int_0^3 |4x-10| dx$

7. Evaluate:

a) $\int_0^1 \frac{e^{3x+1}}{3} dx$

b) $\int_1^3 (x+2)(x^2-2x+3) dx$

8. Evaluate $\int_{-1}^3 (x-3)(x^2+5x) dx$.

9. Evaluate $\int_0^2 x^{\frac{1}{3}}(x^3-2x^2+3) dx$.

10. Solve $\int \sqrt{x}(x^3-2x) dx$.

11. Find $\int \frac{\sqrt[3]{x} + 3x^2 - 4}{\sqrt[3]{x}} dx$.

12. Evaluate $\int_1^4 \sqrt[3]{x} \left(x^3 + \frac{1}{\sqrt[3]{x}}\right) dx$.

13. Evaluate $\int \frac{(2-x)^2 - x}{x^3} dx$.

15. Find $\int \frac{3 \cos^4 \theta - 5 \cos \theta}{\cos^3 \theta} d\theta$.

16. Find $\int \frac{(3 + \sqrt{x})^2}{3x} dx$.

17. Find $\int t^{\frac{1}{5}}(t-3)^2 dt$.

18. Evaluate $\int_3^4 \left(2x + \frac{1}{x}\right)^2 dx$.

19. Find $\int_{-1}^1 \frac{3x^{-5} + 2x^4}{x^2} dx$.

SECTION B

1. Solve the following integrals by using appropriate substitutions:

a) $\int \frac{4x}{3x^2 - 1} dx$

b) $\int \sin 2x [\cos(2x) - 1]^3 dx$

2. Solve the following integrals by using appropriate substitutions:

a) $\int \frac{\sin(2x)}{\cos(2x) + 5} dx$

b) $\int \theta(\theta - 2)^3 d\theta$

3. Solve the integral $\int \tan(4\theta) d\theta$ using suitable substitution.

4. Solve the integral $\int (3 - \sin^2 \theta) \cos \theta d\theta$ using suitable substitution.

6. Solve the following integrals using suitable substitutions.

a) $\int \frac{3x}{(1-2x)^5} dx$

b) $\int \frac{5 \sec^2 x}{(2 + \tan x)^2} dx$

7. a) Find $\int (\cos^2 x + \cos^4 x) \sin x dx$.

- b) Evaluate $\int x^2(x^3 - 6)^7 dx$ using the substitution $u = x^3 - 6$.

8. Evaluate each of the following integrals using appropriate substitutions.

a) $\int_2^4 \frac{5x}{(x-3)^4} dx$

b) $\int \sin 4x e^{\cos 4x} dx$

12. Evaluate each of the following integrals using appropriate substitutions.

a) $\int x(x-5)^5 dx$

b) $\int \frac{2 \sin 5x}{(4 + \cos 5x)^2} dx$

13. Evaluate each of the following integrals using the appropriate substitutions.

a) $\int \frac{(4 + \ln 3x)^5}{x} dx$

b) $\int_0^{\frac{\pi}{6}} e^{5 \tan(2x)} \sec^2(2x) dx$

14. Evaluate each of the following integrals using the appropriate substitutions.

a) $\int (5 + \cos^5 x) \sin x dx$

b) $\int \frac{2}{7(2x+3)^5} dx$

15. Evaluate each of the following integrals using the appropriate substitutions.

a) $\int (3 - \cos^2 \theta)^2 \sin \theta d\theta$

b) $\int \frac{2x}{\sqrt{3x+1}} dx$

SECTION C

1. Evaluate each of the following integrals.

(a) $\int \frac{x-1}{x} dx$

(b) $\int \frac{x}{x-1} dx$

(c) $\int \frac{3x^2}{x-2} dx$

(d) $\int \frac{x^3+1}{x+2} dx$

(e) $\int \frac{4x+1}{2x^2+x-3} dx$

(f) $\int \frac{x+2}{x^2+4x+1} dx$

2. Evaluate each of the following integrals.

(a) $\int \frac{1}{(x+2)(x-4)} dx$

(b) $\int \frac{3}{(x-3)(x-7)} dx$

(c) $\int \frac{3x}{(2x+3)(x+1)} dx$

(d) $\int \frac{3x+1}{(x-2)(x+3)} dx$

(e) $\int \frac{x}{x^2+x-6} dx$

(f) $\int \frac{3x}{(x+2)(x-3)(x+1)} dx$

(g) $\int \frac{3}{x^2+5x+6} dx$

(h) $\int \frac{3x+2}{(x^2-4)(x^2-9)} dx$

Answer:

SECTION A

1. 6.5296
2. a) 114 b) 2
3. $-\frac{1}{x} + \frac{4}{\sqrt{x}} + \ln|x| + c$
4. $\frac{4}{5}x^{\frac{5}{4}} - \frac{8}{9}x^{\frac{9}{4}} + \frac{4}{13}x^{\frac{13}{4}} + c$
5. 29.1776
6. a) 1 b) 12
7. a) 5.7644 b) 28
8. -64/3
9. 4.274
10. $\frac{2}{9}x^{\frac{9}{2}} - \frac{4}{5}x^{\frac{5}{2}} + c$
11. $x + \frac{9}{8}x^{\frac{8}{3}} - 6x^{\frac{2}{3}} + c$
12. 96.548
13. $-\frac{2}{x^2} + \frac{5}{x} + \ln|x| + c$
14. -
15. $3 \sin \theta - 5 \tan \theta + c$
16. $3 \ln|x| + 4\sqrt{x} + \frac{1}{3}x + c$
17. $\frac{5}{16}t^{\frac{16}{5}} - \frac{30}{11}t^{\frac{11}{5}} + \frac{15}{2}t^{\frac{6}{5}} + c$
18. $\frac{641}{12}$
19. 4/3

SECTION B

1. a) $\frac{2}{3}\ln|3x^2 - 1| + c$
b) $-\frac{1}{8}[(\cos 2x) - 1]^4 + c$
2. a) $-\frac{1}{2}\ln|\cos 2x + 5| + c$
b) $\frac{(\theta-2)^5}{5} + \frac{(\theta-2)^4}{2} + c$
3. $-\frac{1}{4}\ln|\cos 4\theta| + c$
4. $3 \sin \theta - \frac{1}{3}\sin^3 \theta + c$
6. a) $\frac{3}{16(1-2x)^4} - \frac{1}{4(1-2x)^3} + c$
b) $-\frac{5}{2+\tan x} + c$
7. a) $-\frac{\cos^3 x}{3} - \frac{\cos^5 x}{5} + c$

7. b) $\frac{1}{24}(x^3 - 6)^8 + c$

8a) -10 b) $-\frac{1}{4}e^{\cos 4x} + c$

12a) $\frac{1}{7}(x-5)^7 + \frac{5}{6}(x-5)^6 + c$

12b) $\frac{2}{5(4+\cos 5x)} + c$

13a) $\frac{1}{6}(4 + \ln 3x)^6 + c$ 13b) 14.7413

14a) $-5 \cos x - \frac{1}{6}\cos^6 x + c$

14b) $-\frac{1}{28(2x+3)^4} + c$

15a) $-9 \cos \theta + 2 \cos^3 \theta - \frac{1}{5}\cos^5 \theta + c$

15b) $\frac{4}{27}(3x+1)^{\frac{3}{2}} - \frac{4}{9}(3x+1)^{\frac{1}{2}} + c$

SECTION C

DIY